FACULDADE DE DESIGN,
TECNOLOGIA E COMUNICAÇÃO
DA UNIVERSIDADE EUROPEIA

Universidade Europeia

MASTER'S IN CREATIVE COMPUTING
AND ARTIFICIAL INTELLIGENCE

EEG-Based Remote Car Control with Emotiv EPOC X:

An Open-Source Python Framework and Experimental Evaluation

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Prof. Dr. João A. Dias

IADE, Faculdade de Design, Tecnologia e Comunicação
da Universidade Europeia

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Universidade Europeia

2026

DUMILDE JOSÉ DE MATOS

EEG-BASED REMOTE CAR CONTROL WITH EMOTIV EPOC X: AN OPEN-SOURCE PYTHON FRAMEWORK AND EXPERIMENTAL EVALUATION

MOD-195.IADEV02; 07-07-2023

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EEG-BASED REMOTE CAR CONTROL WITH EMOTIV EPOC X: AN OPEN-SOURCE PYTHON FRAMEWORK AND EXPERIMENTAL EVALUATION

Dissertação apresentada ao IADE - Faculdade de Design, Tecnologia e Comunicação da Universidade Europeia, para cumprimento dos requisitos necessários à obtenção do grau de Mestre em Creative Computing and Artificial Intelligence realizada sob a orientação científica da Doutora Pamela Teubig, professora assistente do *IADE* e do Doutor João A. Dias, professor do IADE.

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MOD-195.IADEV02; 07-07-2023

Keywords *[Keywords to add.]*

abstract *[Text to write.]*

Glossary

Term *[Definition to write.]*

List of Abbreviations

ABBR *[Meaning to write.]*

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[Text to write.]

1.1 Motivation

[Text to write.]

1.2 Objective

[Text to write.]

1.2.1 Main Objective

[Text to write.]

1.2.2 Secondary Objectives

[Text to write.]

1.3 Hypothesis

[Text to write.]

1.3.1 Improved depth accuracy with optical principles for lens-based imaging

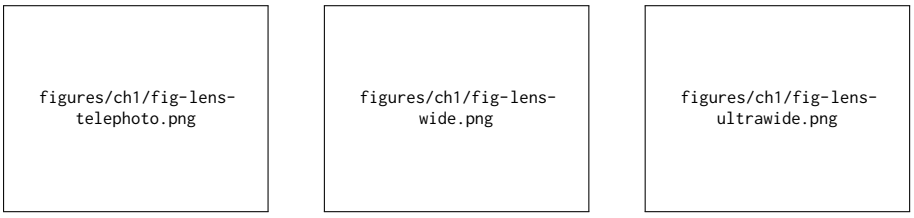
[Text to write.]

1.3.2 Anisotropic Gaussian optimization

[Text to write.]

1.3.3 To put it simply

[Text to write.]



(a) Telephoto lens (b) Wide lens (c) Ultra-wide lens

Figure 1: Comparison of images captured using telephoto, wide, and ultra-wide lenses.

2 3D Scene Reconstruction: understanding photogrammetry, novel view synthesis and stereoscopy

[Text to write.]

2.1 The breakthrough in Côte d'Azur

[Text to write.]



Figure 2: 3D point cloud (bottom) generated from images (top). From *3D Digitization using Structure from Motion* (Haro et al., 2012).

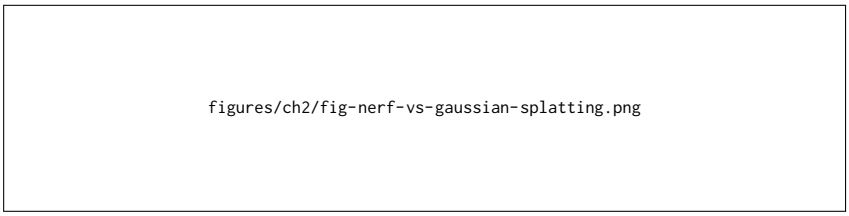


Figure 3: A conceptual comparison between NeRFs and Gaussian Splatting, originally presented in *A Comprehensive Overview of Gaussian Splatting* (Yurkova, 2024).

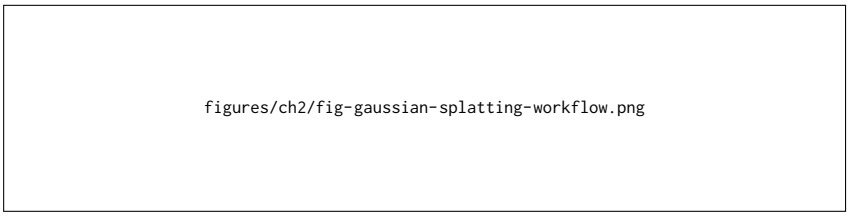


Figure 4: Representation of a traditional Gaussian Splatting workflow, originally presented in *3D Gaussian Splatting for Real-Time Radiance Field Rendering* (Kerbl et al., 2023).

2.2 3DGS-LiDAR, a hardware assisted approach for Gaussian Splatting model improvements

[Text to write.]

2.2.1 How much extra data is actually helpful?

[Text to write.]

2.2.2 Optimizing image overlap

[Text to write.]

2.2.3 Merging Point Clouds

[Text to write.]

2.2.4 Improvements in final output

[Text to write.]

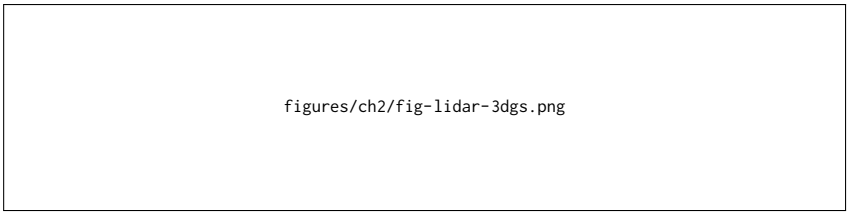


Figure 5: Originally from *LiDAR-3DGS: LiDAR Reinforced 3D Gaussian Splatting for Multimodal Radiance Field Rendering* (Lim et al., 2024).

2.3 Revisiting stereoscopy: a multi-lens approach for recreating a 3D scene

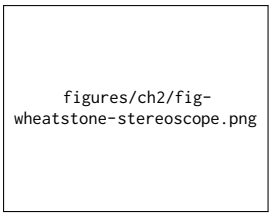
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2.4 COLMAP and OpenCV as relevant tools and libraries

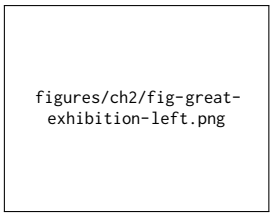
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2.5 Lens distortions and important photography concepts

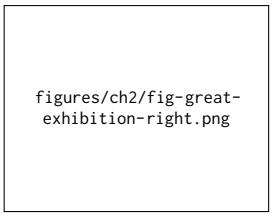
[Text to write.]



(a) Wheatstone's stereoscope



(b) 1851 Great Exhibition - L



(c) 1851 Great Exhibition - R

Figure 6: First stereoscopic device, with details of the pictures used for left eye (L) and right eye (R) viewing. Reproduced from the King's College London archive.

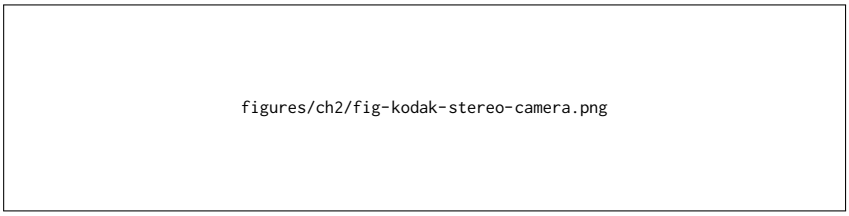


Figure 7: Example of a Kodak Stereo Camera, a 35mm stereoscopic film camera manufactured during 1954 and 1959. Photography by John Alan Elson, licensed under CC BY-SA 4.0.

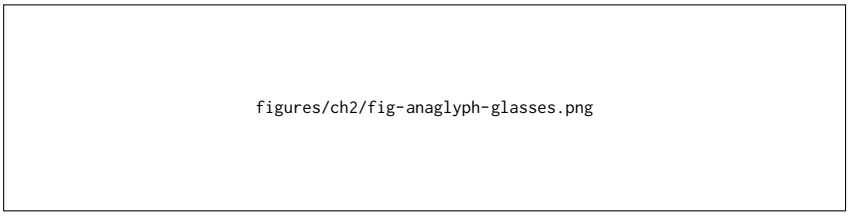


Figure 8: Passive anaglyph 3D glasses mechanics illustration sourced from *How 3D Glasses Work?* (Brain, 2000).

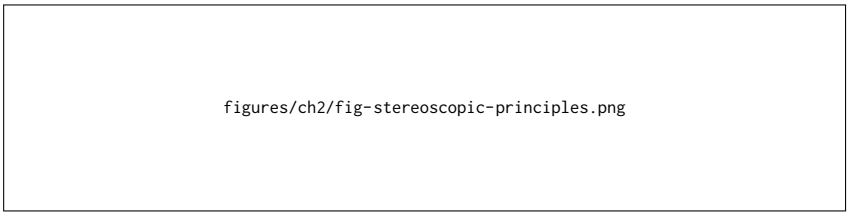


Figure 9: Sourced from *Basic Principles of Stereoscopic 3D* (Reeve and Flock, 2010)

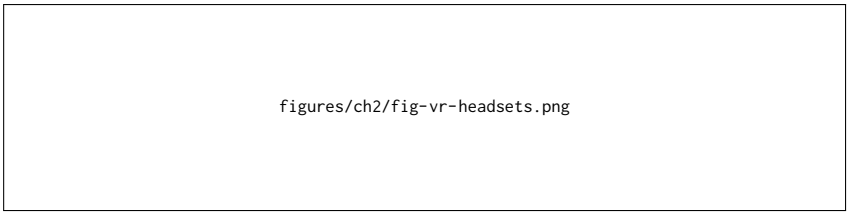


Figure 10: Sourced from *What are VR headsets?* (Interaction Design Foundation - IxDF, 2024)

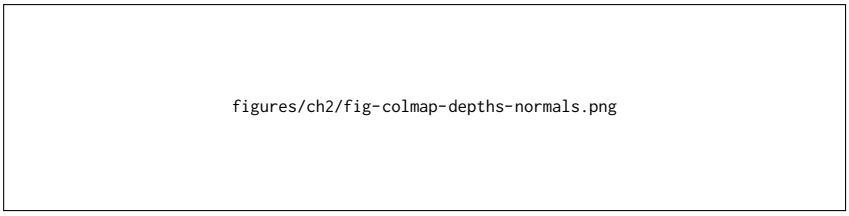
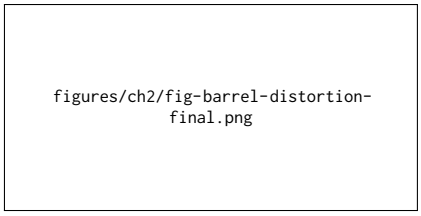


Figure 11: Images with filtered depths and normals for crowd-sourced images, after Colmap Dense Reconstruction. Presented originally in *Pixelwise View Selection for Unstructured Multi-View Stereo* (Schönberger et al., 2016).



(a) Processing barrel distortion on source image



(b) Final image after calibration

Figure 12: Image undistortion process using OpenCV, presented originally in *OpenCV Camera Calibration* (Bradski, 2000).



Figure 13: Example of lens distortion originally presented on *Learning OpenCV* (Bradski and Kaehler, 2011)

3 Multi-lens 3DGS: an approach to power 3D scans leveraging wide and ultrawide lenses with data collected from consumer-grade devices

[Text to write.]

3.1 OpenSplat as the chosen base code

[Text to write.]

3.2 Choosing a proper dataset: ARKitScenes

[Text to write.]

3.2.1 Isolating scenes images

[Text to write.]

```
# Listing 1: replace with the script.
```

Listing 1: Python script for downsampling the raw wide folder so that it has the same image count as the ultrawide folder

3.2.2 Ultrawide preprocessing: undistortion and cropping

[Text to write.]

```
# Listing 2: replace with the script.
```

Listing 2: Python script for fisheye undistortion using OpenCV

```
# Listing 3: replace with the script.
```

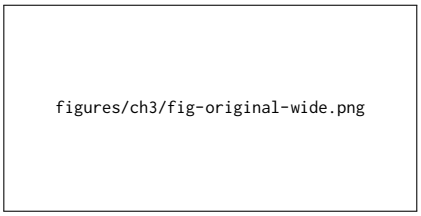
Listing 3: Developed python script used for cropping undistorted ultrawide images

3.2.3 One scene, three scenarios

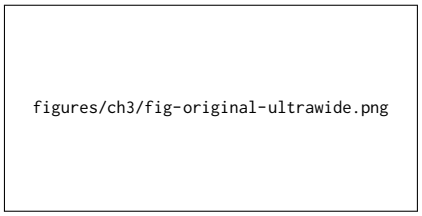
[Text to write.]

3.3 Generating a sparse point cloud

[Text to write.]



(a) Original wide image

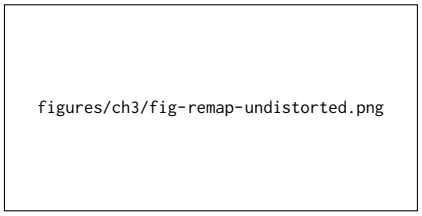


(b) Original ultrawide image

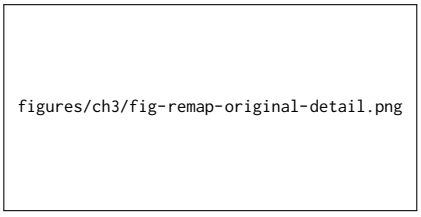
Figure 14: Comparison between wide and ultrawide images, showcasing the difference in Field of View between the lenses.



(a) Original ultrawide image



(b) Undistorted ultrawide image

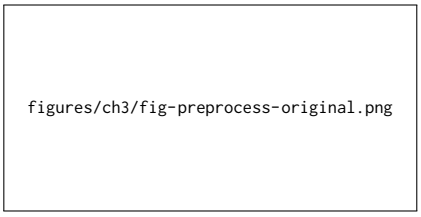


(c) Detail of original ultrawide image

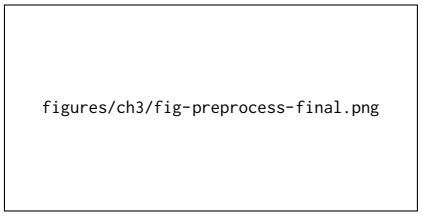


(d) Detail of undistorted ultrawide

Figure 15: Remap undistortion process, with details for both original and undistorted images. In images c and d, there is change in the lines formed by the blinds in the window, with those bending to a more intensive degree on image c than on image d.



(a) Original ultrawide image



(b) Preprocessed ultrawide image

Figure 16: Comparison between original and preprocessed ultrawide images

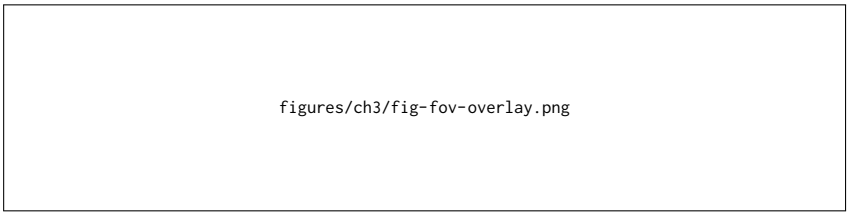


Figure 17: Overlaying comparison between final preprocessed ultrawide image and original wide image, with the overlapping FOV applied with a difference blend mode and the extra FOV still present after undistortion and cropping represented normally, as a frame to the wide image.

3.3.1 Feature extraction

[Text to write.]

3.3.2 Sequential Feature Matching

[Text to write.]

3.3.3 Reconstruction (Multi-Model)

[Text to write.]

3.4 Batch running OpenSplat with custom metric log

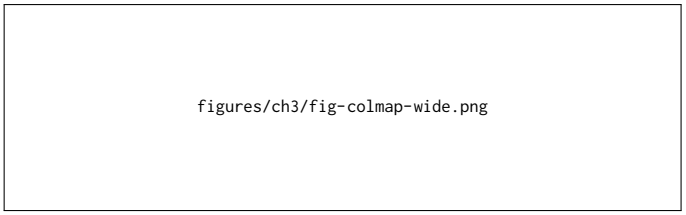
[Text to write.]

```
1 // Listing 4: replace with the code.
```

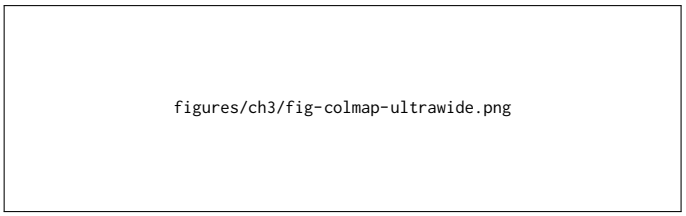
Listing 4: C++ function for logging SSIM-weighted loss for each step on console

```
1 # Listing 5: replace with the script.
```

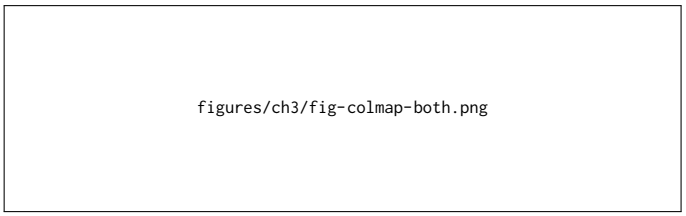
Listing 5: Bash script for processing scenes and recording metrics



(a) COLMAP reconstruction using only wide lens images (Scene 41048085)



(b) COLMAP reconstruction using preprocessed ultrawide lens images (Scene 41048085)



(c) COLMAP reconstruction using both wide and preprocessed ultrawide images (Scene 41048085)

Figure 18: Qualitative comparison of COLMAP reconstructions under different input conditions for Scene 41048085.

4 Methodology

[Text to write.]

4.1 Performance Metrics

[Text to write.]

4.1.1 SSIM

[Text to write.]

4.1.2 MSE and PSNR

[Text to write.]

4.1.3 Time

[Text to write.]

5 Results

[Text to write.]

5.1 SSIM Analysis

[Text to write.]

Table 1: SSIM per Scene and Input Type

Scene	Both	Just Wide	Just Ultrawide	Best Output
[...]	*[...]*	*[...]*	*[...]*	*[...]*
Mean	*[...]*	*[...]*	*[...]*	
Std Dev	*[...]*	*[...]*	*[...]*	

5.2 PSNR Analysis

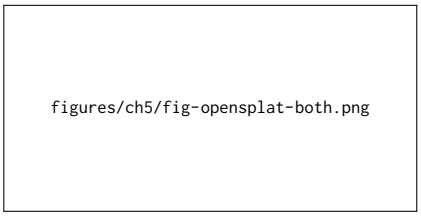
[Text to write.]

Table 2: MSE per Scene and Input Type with Best Output

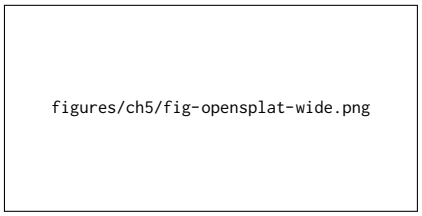
Scene	Both	Just Wide	Just Ultrawide	Best Output
[...]	*[...]*	*[...]*	*[...]*	*[...]*
Mean	*[...]*	*[...]*	*[...]*	
Std Dev	*[...]*	*[...]*	*[...]*	

5.3 The scores and the constraints

[Text to write.]



(a) OpenSplat output for both lenses, showing in detail the tunneling effect



(b) OpenSplat output for just wide lenses, showing in detail straight lines

Figure 19: Side-by-side comparison of OpenSplat reconstructions of scene **41048113**

6 Conclusion

[Text to write.]

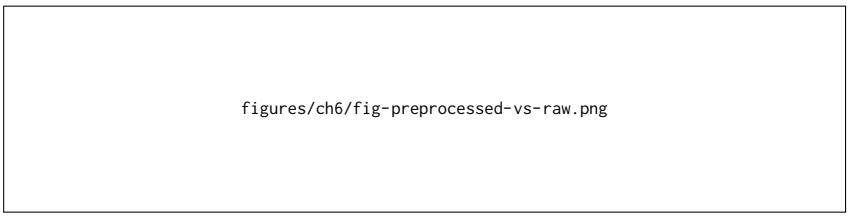
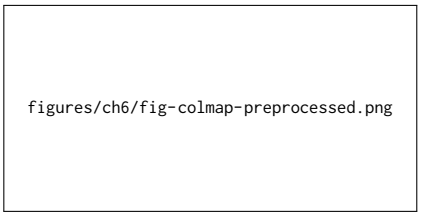
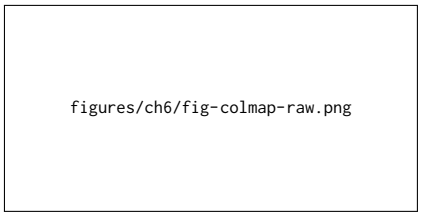


Figure 20: Comparison between preprocessed and raw ultrawide images.



(a) COLMAP with preprocessed ultrawide



(b) COLMAP with raw ultrawide

Figure 21: Side-by-side comparison of COLMAP reconstructions

References

Besl, P. and McKay, N. D. (1992). A method for registration of 3-d shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256.

Annex

```
1 // Listing 6: replace with the code.
```

Listing 6: Tweaked model.cpp file used in study

Faculdade de Design,
Tecnologia e Comunicação
Universidade Europeia

2026

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